

Counting Unlabeled Chordal Graphs by Equivariant Evaporation

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June 2026

Abstract

We compute the number of unlabeled chordal graphs on n vertices, both the total count ([OEIS A048193](#)) and the connected count ([OEIS A048192](#)), extending two sequences whose published values had remained at $n = 15$. The method is a Pólya–Burnside enumeration: the number of unlabeled graphs in a class closed under relabeling is the average over S_n of the number of labeled graphs fixed by each permutation. The technical core is the evaluation, for an arbitrary permutation π , of the number of π -invariant labeled chordal graphs. We give a dynamic program for this quantity that lifts the evaporation-based labeled chordal counting of Hébert-Johnson, Lokshtanov and Vigoda to the equivariant setting. Its central structural ingredient is a *divisor-bundle decomposition*: when a connected piece spans a cyclic orbit of size c , it forms, for each divisor $d \mid c$, a d -fold bundle whose constituent is an object of the same kind in the cyclic world of order c/d , computed by the same program recursively. We prove the decomposition and the correctness of the resulting recurrences, and we prove that the full Burnside computation runs in sub-exponential time $n^{O(\sqrt{n})}$. We report the new terms through $n = 18$ and describe four independent validations, including exact agreement with all previously known values of both sequences and an Euler-transform consistency check.

1 Introduction

A graph is *chordal* if it has no induced cycle of length at least four. Chordal graphs are a central class in algorithmic graph theory, characterized by perfect elimination orderings and by tree decompositions whose width equals the clique number minus one. Two integer sequences count them up to isomorphism:

$$\begin{aligned} \text{A048193}(n) &= \#\{\text{chordal graphs on } n \text{ nodes}\}, \\ \text{A048192}(n) &= \#\{\text{connected chordal graphs on } n \text{ nodes}\}, \end{aligned}$$

both up to isomorphism. Their published data extended only to $n = 15$.

For the *labeled* count, Hébert-Johnson, Lokshtanov and Vigoda [1] gave a polynomial-time dynamic program based on an *evaporation* process that repeatedly deletes all simplicial vertices. Counting graphs *up to isomorphism*, however, is governed by Burnside’s lemma and requires counting labeled chordal graphs invariant under each permutation of the vertices. Hébert-Johnson and Lokshtanov [2] proved that counting labeled chordal graphs with a prescribed automorphism is fixed-parameter tractable in the number of moved points μ , with a running time of $O(2^{7\mu}n^9)$; in the regime relevant to the Burnside sum, where μ may be as large as n , that bound is single-exponential in n , and no exact unlabeled enumeration was carried out there.

Contributions. Building on the labeled evaporation dynamic program of [1], we give a concrete equivariant dynamic program for the number $\text{fix}(\pi)$ of π -invariant labeled chordal graphs, and use it to compute A048192 and A048193 beyond the previous frontier. Our results are:

1. A *divisor-bundle* structure theorem (the Component-Orbit Decomposition, Theorem 1) describing how connected pieces of a π -invariant graph decompose, and reducing each equivariant recurrence to its labeled counterpart of [1] (Section 5.3).
2. A correctness theorem (Theorem 3) for the assembled program, conditional only on the labeled recurrences of [1].
3. A running-time theorem (Theorem 4): the entire computation of A048192(n) and A048193(n) runs in sub-exponential time $n^{O(\sqrt{n})} = e^{O(\sqrt{n} \log n)}$, in contrast to the single-exponential worst case of the general parameterized bound.
4. The new values through $n = 18$ (Section 7), with four independent validations.

2 Preliminaries

2.1 The Burnside reduction

Let $\mathcal{H}$ be a class of graphs on a fixed vertex set that is closed under relabeling (chordality is such a property). The number of isomorphism classes of $\mathcal{H}$ -graphs on n vertices is, by the Cauchy–Frobenius–Burnside lemma,

$$a(n) = \frac{1}{n!} \sum_{\pi \in S_n} \text{fix}(\pi) = \sum_{\lambda \vdash n} \frac{\text{fix}(\lambda)}{z_\lambda}, \quad z_\lambda = \prod_{i \geq 1} i^{m_i} m_i!, \quad (1)$$

where $\text{fix}(\pi)$ is the number of $\mathcal{H}$ -graphs invariant under π (i.e. with $\pi \in \text{Aut}(G)$), a quantity that depends only on the cycle type $\lambda = (1^{m_1} 2^{m_2} \dots)$ of π , and $z_\lambda = n! / |\{\pi : \text{type } \lambda\}|$. The connected count obeys the same identity with fix replaced by cfix , the number of *connected* invariant graphs, since S_n acts on connected labeled graphs as well. Thus everything reduces to computing $\text{fix}(\lambda)$ and $\text{cfix}(\lambda)$ for each cycle type λ .

Throughout, fix $\pi \in S_V$ with V the vertex set, $|V| = n$, and let $N = \text{lcm}$ of the cycle lengths $= \text{ord}(\pi)$. The orbits of π partition V ; an orbit of size c is a *c-cycle*, on which π acts as a single cycle. A part of size c in λ is such an orbit.

2.2 Evaporation

Definition 1 (Evaporation with an exception clique; [1]). A vertex v of a graph G is *simplicial* if its neighborhood $N_G(v)$ is a clique. Given a clique $X \subseteq V(G)$, set $G_0 = G$ and, for $t \geq 1$, let L_t be the set of simplicial vertices of G_{t-1} not in X and put $G_t = G_{t-1} \setminus L_t$. For chordal G this terminates with $V(G_t) = X$; the sequence $L_1, \dots, L_T$ is the *evaporation of G with exception X* , G *evaporates at time T* , the *evaporation time* of $v \notin X$ is the index t with $v \in L_t$, and $L_G(X) := L_T$ is the last layer.

We write $\tau(v)$ for the evaporation time of v .

3 The orbit-level process and its structure

We run the evaporation process on a π -invariant chordal graph with a π -invariant exception clique. The following lemma is what makes an equivariant dynamic program well-defined: the process acts

on π -orbits, not on individual vertices.

Lemma 1 (Orbit-evaporation). *Let G be chordal with $\pi \in \text{Aut}(G)$, and let the exception clique X be π -invariant. Then $\pi(L_t) = L_t$ and $\pi \in \text{Aut}(G_t)$ for every t . Consequently τ is constant on each π -orbit: every orbit evaporates in a single layer.*

Proof. For any π -invariant graph H one has $N_H(\pi v) = \pi(N_H(v))$, and π maps cliques to cliques; hence v is simplicial in H iff πv is. We induct on t . Base: $G_0 = G$ is π -invariant. Step: if $\pi \in \text{Aut}(G_{t-1})$ then simpliciality in G_{t-1} is π -stable and X is π -invariant, so $\pi(L_t) = L_t$; deleting the π -invariant set L_t leaves π an adjacency-preserving bijection of $V(G_t)$, i.e. $\pi \in \text{Aut}(G_t)$. Since $\pi(L_t) = L_t$, each π -orbit is contained in or disjoint from L_t , so τ is constant on orbits. $\square$

A connected component need not be π -invariant; rather π permutes the components. The orbit of a component is a *bundle*.

Lemma 2 (Component-orbit bundle). *Let G be π -invariant and K a connected component of G , with π -orbit $\mathcal{B} = \{K, \pi K, \dots, \pi^{d-1}K\}$ of size d . Then:*

- (i) $d \mid N$ and $\text{Stab}(K) = \langle \pi^d \rangle$, so K is π^d -invariant;
- (ii) the d components of $\mathcal{B}$ are pairwise isomorphic and $\bigcup_j \pi^j K$ is π -invariant.

Proof. $\langle \pi \rangle$ is cyclic of order N . By orbit-stabilizer $[\langle \pi \rangle : \text{Stab}(K)] = d$, so $|\text{Stab}(K)| = N/d$ and $d \mid N$. A cyclic group has a unique subgroup of each order dividing N ; the one of order N/d is $\langle \pi^d \rangle$, giving (i). Each π^j restricts to an isomorphism $K \rightarrow \pi^j K$, and a union of a $\langle \pi \rangle$ -orbit is π -invariant, giving (ii). $\square$

Lemma 3 (Block decomposition). *Let O be a π -orbit of size s and $H = \langle \pi^d \rangle$ for $d \mid N$. Then H acts on O with exactly $\gcd(d, s)$ orbits (blocks), each of size $s/\gcd(d, s)$. Any π^d -invariant set meets O in a union of whole blocks.*

Proof. Identify $O \cong \mathbb{Z}/s\mathbb{Z}$ with $\pi|_O$ translation by 1, so $\pi^d|_O$ is translation by d . The orbits of $\langle d \rangle \leq \mathbb{Z}/s\mathbb{Z}$ are the cosets of the subgroup generated by d , which has order $s/\gcd(d, s)$; there are $\gcd(d, s)$ of them. A π^d -invariant set is a union of such orbits. $\square$

Lemma 4 (Recursive sub-world). *With $K, \mathcal{B}, d$ as in Lemma 2, the component K is a connected chordal graph invariant under the cyclic group $H = \langle \pi^d \rangle$ of order N/d . Every π -orbit O of size s that K meets contributes to K a union of blocks of size $s/\gcd(d, s)$. Moreover, if O has size c with $d \mid c$ and is one of the orbits spanned by $\mathcal{B}$, then K meets O in exactly one block, a single H -orbit of size c/d . Hence K is an object of the same type on a strictly smaller cycle type, and the bundle structure recurses, terminating at $d = N$ in the labeled world.*

Proof. K is H -invariant (Lemma 2(i)) and H is cyclic, so by Lemma 3 $K \cap O$ is a union of blocks of size $s/\gcd(d, s)$. For a spanned O with $d \mid c$ we have $\gcd(d, c) = d$, so O has exactly d blocks, each of size c/d . The bundle has period d , so its d components are distinct and partition $\bigcup_j \pi^j K \supseteq O$. As $K \cap O \neq \emptyset$ and $\pi O = O$, each translate meets O in $\pi^j(K \cap O) \neq \emptyset$; these d sets are pairwise disjoint and cover O , and π^j is a bijection, so $d|K \cap O| = |O| = c$ and $|K \cap O| = c/d$. Since $K \cap O$ is a union of size- (c/d) blocks of total size c/d , it is exactly one block. The spanned orbit of size c thus becomes a sub-orbit of size $c/d < c$ (as $d > 1$), so the recursion is well-founded and ends when $d = N$, where $H = \{\text{id}\}$ and K is an unconstrained labeled graph. $\square$

4 The Component-Orbit Decomposition

For a cyclic world of order M and a cycle type ν , let $g_1^{\langle M \rangle}(t, x, \nu)$ denote the number of configurations consisting of a single connected component-bundle on the new orbits ν , seeing a prescribed boundary, with evaporation time t (Section 5 makes the boundary parameters precise; for the present theorem only its existence is used). The labeled count $\sum_t g_1^{\langle 1 \rangle}(t, 0, 1^k) = \text{A007134}(k)$ is the number of connected labeled chordal graphs on k vertices [1].

Theorem 1 (Component-Orbit Decomposition). *Let μ be a multiset of new orbits and d a positive integer dividing every size occurring in μ . The number of period- d component-bundles spanning exactly μ , seeing a prescribed set of boundary blocks, with evaporation time t , equals*

$$d^{|\mu|-1} g_1^{\langle N/d \rangle}(t, x', \mu/d),$$

where μ/d replaces each size s by s/d , the order- (N/d) world is that of Lemma 4, and the boundary x' records, for each boundary orbit of size s , its $\gcd(d, s)$ blocks of sub-type $s/\gcd(d, s)$. For $d = 1$ the term is the equivariant count $g_1(t, x, \mu)$.

Proof. By Lemma 4 a representative K of such a bundle is, with its incidences to the boundary blocks, a connected configuration in the order- (N/d) world on the sub-orbits μ/d (each spanned size- s orbit met in a single sub-orbit of size s/d) and on the boundary blocks x' ; its evaporation time is t because, by Lemma 1 applied to $\langle \pi \rangle \supseteq \langle \pi^d \rangle$, τ is constant on the $\langle \pi^d \rangle$ -orbits that are the sub-vertices of K , so the sub-world evaporation has the same layers. Conversely the map $K \mapsto \bigcup_j \pi^j K$ sends each such sub-configuration to a period- d bundle (a disjoint union of the chordal $\pi^j K$, π -invariant by Lemma 2(ii)).

It remains to count the bundles. Fix a distinguished spanned orbit O_1 and let w_1 be its least vertex. We exhibit a bijection

$$\{\text{period-}d \text{ bundles on } \mu\} \longleftrightarrow \{\text{order-}(N/d) \text{ sub-configurations}\} \times \{1, \dots, d\}^{|\mu|-1}.$$

Forward map. Given a bundle $\mathcal{B}$, let K be the component containing w_1 (well-defined: w_1 lies in exactly one of the d blocks of O_1 , hence in exactly one component, since the translates partition O_1 by the argument of Lemma 4). Record the isomorphism type of K as an order- (N/d) sub-configuration, and the alignment $a = (a_i)_{i \geq 2}$, where $a_i \in \{1, \dots, d\}$ names which block of O_i lies in K (each O_i contributes exactly one block to K by Lemma 4); the block of O_1 in K is fixed to be the one containing w_1 .

The period is exactly d . For $0 < j < d$, $\pi^j K$ meets O_1 in $\pi^j(K \cap O_1)$, a block distinct from $K \cap O_1$ (the d blocks of O_1 are permuted freely by π , as in Lemma 4); hence $\pi^j K \neq K$, so $\mathcal{B}$ has exactly d distinct components and the forward map lands in period- d bundles regardless of any abstract automorphism of K . This rules out the only possible source of over-collapse — a representative with internal periodic symmetry — because the symmetry that would matter, $\pi^j K = K$, is forbidden by the block displacement.

Injectivity. The type of K and the alignment determine the vertex set and edges of K exactly (the alignment fixes which concrete blocks of each O_i are in K , and the type fixes the adjacencies among them and to the boundary blocks); K then determines $\mathcal{B} = \{K, \pi K, \dots, \pi^{d-1} K\}$.

Surjectivity and freeness. Conversely, any sub-configuration K and alignment $a \in \{1, \dots, d\}^{|\mu|-1}$ yields a bundle: place K on the blocks named by a (and the w_1 -block of O_1), set $\mathcal{B} = \bigcup_j \pi^j K$. This is π -invariant (Lemma 2), chordal (disjoint union of chordal pieces), of period d (above), and spans μ with the prescribed block-seeing. The alignment choices are unconstrained: changing a_i relabels

the order- (N/d) sub-orbit of K on O_i to a different block of O_i , a graph isomorphism that preserves connectivity and all boundary incidences (those are recorded against boundary *blocks*, which are fixed and shared by all components, hence unaffected by the new-orbit alignment). Thus the boundary-seeing data and the alignment are independent, and there are exactly $d^{|\mu|-1}$ alignments.

Therefore the number of bundles is $d^{|\mu|-1}$ times the number of order- (N/d) sub-configurations of the prescribed type, which is $g_1^{\langle N/d \rangle}(t, x', \mu/d)$. $\square$

Summing over admissible periods gives the count of all bundles spanning μ :

$$\text{COD}(t, x, \mu) = \underbrace{g_1(t, x, \mu)}_{d=1} + \sum_{\substack{d > 1 \\ d | \gcd(\mu)}} d^{|\mu|-1} g_1^{\langle N/d \rangle}(t, x', \mu/d), \quad (2)$$

where the boundary parameter x' and the admissible boundary set are specialized to the condition in force (Section 5.3).

5 The dynamic program and its correctness

5.1 State variables and the block decomposition

Fix the distinct orbit sizes $S = (s_1 < \dots < s_r)$ of π . All state variables are integer vectors indexed by S :

$$x, k, z, l \in \mathbb{Z}_{\geq 0}^r, \quad z \leq x \text{ (componentwise)}.$$

Here x_i is the number of *boundary* orbits of size s_i (their union X is a clique); k_i is the number of *new* orbits of size s_i ; l_i (for the f -functions) is the number of orbits in the last evaporation layer $L_G(X)$; and z is an inclusion–exclusion sub-boundary, specifying — under a fixed linear order on the boundary orbits — a downward-closed sub-clique used to express “the piece reaches beyond z .” A vertex count is recovered as $|X| = \sum_i s_i x_i$, etc. There is no separate “seeing” state variable: which boundary a piece sees is summed over inside the recurrences (below), exactly as in the labeled case, where it is the variable $x' \leq x$.

The one genuinely new ingredient is how a *bundle* interacts with the boundary. By Lemma 3, a period- d bundle does not see boundary *orbits* but boundary *blocks*: a boundary orbit of size s_i presents $\gcd(d, s_i)$ blocks of sub-type $\tau_i := s_i / \gcd(d, s_i)$. We therefore define, for each $d \mid N$, the *block profile* of the boundary x as the vector indexed by the distinct sub-types $\{\tau_i\}$:

$$\beta_d(x)_\tau = \sum_{i: \tau_i = \tau} \gcd(d, s_i) x_i \quad (\text{number of blocks of sub-type } \tau), \quad (3)$$

and the associated boundary generating function in formal variables $\{v_\tau\}$,

$$\Gamma_d(x) = \prod_{i=1}^r (1 + v_{\tau_i})^{\gcd(d, s_i) x_i}, \quad \Gamma_d^{\text{cov}}(x) = \prod_{i=1}^r \left((1 + v_{\tau_i})^{\gcd(d, s_i)} - 1 \right)^{x_i}. \quad (4)$$

The coefficient of $\prod_\tau v_\tau^{a_\tau}$ in $\Gamma_d(x)$ is the number of ways a bundle’s sub-component can see exactly a_τ blocks of each sub-type; Γ_d^{cov} is the same restricted to seeing at least one block of every boundary orbit (“covers all of X ”). The parenthetical is exact, not merely suggestive: by Lemma 3 the quotient $\langle \pi \rangle / \langle \pi^d \rangle \cong \mathbb{Z} / \gcd(d, s_i) \mathbb{Z}$ permutes the $\gcd(d, s_i)$ blocks of each boundary orbit O_i transitively, so the full period- d bundle $K \cup \pi K \cup \dots \cup \pi^{d-1} K$ is adjacent to all of O_i exactly when its representative K meets at least one block of O_i (the d translates sweep that block across the whole

orbit). Thus Γ_d^{cov} counts component-orbits that, taken as a whole, see all of X , even when no single constituent component does. These are the equivariant analogues of the labeled coefficients $\binom{x}{x'}$ and $\binom{x}{x}$, and the sub-type vector $x' = (a_\tau)_\tau$ is the boundary argument of the sub-world function in Theorem 1. With $d = 1$ every block is a whole orbit ($\tau_i = s_i$, $\gcd = 1$) and $\Gamma_1(x) = \prod_i (1 + v_{s_i})^{x_i}$ reduces to choosing a sub-vector $x' \leq x$ of boundary orbits, recovering the labeled situation.

5.2 The counter functions

We use the eight functions of [1, Def. 3.3], read on the orbit-level evaporation of Lemma 1 over the state variables above; “component” means component-bundle. Briefly, for a π -invariant chordal G with X a clique:

$g(t, x, k, z)$	every component of $G \setminus X$ sees $X \setminus [z]$; evaporates in time $\leq t$
$\tilde{g}(t, x, k, z)$	as g , with every component evaporating at time exactly t
$\hat{g}(t, x, k, z)$	as $\tilde{g}$, with no component seeing all of X
$g_1(t, x, k)$	as $\tilde{g}$, exactly one component, seeing all of X
$g_2(t, x, k)$	as g_1 but ≥ 2 components, each seeing all of X
$f(t, x, l, k)$	connected; evaporates at exactly t ; $L_G(X) = [x+1, x+l]$; $X \cup L_G(X)$ a clique
$\tilde{f}(t, x, l, k)$	as f , components of $G \setminus (X \cup L)$ evaporate at exactly $t-1$ (≥ 1)
$\hat{f}(t, x, l, k)$	as $\tilde{f}$, with no component of $G \setminus (X \cup L)$ seeing all of $X \cup L$

The labeled instance ($\pi = \text{id}$) is exactly [1]; there $\text{cfix}_{\text{lab}}(k) = \sum_t g_1(t, 0, k)$.

Theorem 2 (Labeled recurrences; [1], Lemmas 3.4–3.8). *For $\pi = \text{id}$ the eight functions satisfy the recurrences*

$$\begin{aligned}
g_1(t, x, k) &= \sum_{l=1}^k \binom{k}{l} f(t, x, l, k-l), \\
f(t, x, l, k) &= \sum_{k'=1}^k \binom{k}{k'} \tilde{f}(t, x, l, k') g(t-2, x+l, k-k', x), \\
g(t, x, k, z) &= \sum_{k'=0}^k \binom{k}{k'} \tilde{g}(t, x, k', z) g(t-1, x, k-k', z), \\
\tilde{g}(t, x, k, z) &= \sum_{k'=1}^k \sum_{x'=1}^x \left(\binom{x}{x'} - \binom{z}{x'} \right) \binom{k-1}{k'-1} g_1(t, x', k') \tilde{g}(t, x, k-k', z),
\end{aligned}$$

together with the stated recurrences for $\hat{g}$, g_2 , and $\hat{f}$ (Lemma 3.8), all proved by conditioning on the component containing the least non-boundary vertex and recursing on the rest.

5.3 From labeled to equivariant

We lift Theorem 2 to general π ; the resulting eight equivariant recurrences are written explicitly in Appendix A. The four recurrences that compose other functions ($g_1, f, g, \tilde{f}$) lift verbatim: their derivations partition the new vertices by structural role (membership in $L_G(X)$, evaporation time, or seeing-all status), and by Lemma 1 these roles are orbit-stable, so the per-size product of binomials $\binom{k}{\cdot} = \prod_i \binom{k_i}{\cdot}$ replaces the single binomial and each constituent function is its equivariant counterpart. The remaining recurrences condition on a single component; for them we use the following principle.

Proposition 1 (Lift Principle). *Let $F \in \{\tilde{g}, \hat{g}, g_2, \hat{f}\}$, defined by a condition Φ_F on the components of $G \setminus X$ (an evaporation pattern and a boundary-seeing pattern). The equivariant recurrence for F is the recurrence of Theorem 2 for F with the single-component count $g_1(t, x', k')$ replaced by $\text{COD}_{\Phi_F}(t, x', \kappa)$ of (2), restricted to the boundary condition of Φ_F .*

Proof. Fix the distinguished orbit o^* of largest present size (least vertex among ties). By Lemmas 1–2 the components of $G \setminus X$ form component-bundles and o^* lies in a unique one, $\mathcal{C}^*$, spanning some κ with $\kappa_p \geq 1$ at the marked size p . Conditioning on $\mathcal{C}^*$: the orbits joining it are counted by $\binom{k_p-1}{\kappa_p-1} \prod_{i \neq p} \binom{k_i}{\kappa_i}$ (the marked orbit forced in, the rest free); $\mathcal{C}^*$ itself is counted by $\text{COD}_{\Phi_F}(t, x', \kappa)$ by Theorem 1, with the boundary coefficient of Φ_F applied per orbit for the $d = 1$ term and per block for $d > 1$ (Lemma 3); and $G \setminus (X \cup \mathcal{C}^*)$ is an F -configuration on the remaining orbits, counted by the recursive F -term. The three are independent (disjoint vertex sets) and the counts multiply, exactly as in the proof of Theorem 2; the only change is the substitution of COD for the single-component count. $\square$

Proposition 2 (All from connected). *Write the cycle type as $k = (k_1, \dots, k_r)$ over the distinct sizes $s_1 < \dots < s_r$, let $p = \max\{i : k_i > 0\}$, and let $\text{BUNDLE}(\mu)$ be the number of single component-bundles spanning μ , namely*

$$\text{BUNDLE}(\mu) = \text{cfix}(\mu) + \sum_{d > 1, d | \gcd(\mu)} d^{|\mu|-1} \text{cfix}^{\langle N/d \rangle}(\mu/d).$$

Then

$$\text{fix}(k) = \sum_{\substack{\mu \leq k \\ \mu_p \geq 1}} \left[\binom{k_p-1}{\mu_p-1} \prod_{i \neq p} \binom{k_i}{\mu_i} \right] \text{BUNDLE}(\mu) \text{fix}(k - \mu), \quad \text{fix}(\mathbf{0}) = 1.$$

Proof. A π -invariant chordal graph is the disjoint union of its component-bundles (Lemma 2), and chordality is tested componentwise. Distinguish the orbit o^* of size s_p with least vertex; it lies in a unique bundle, spanning μ with $\mu_p \geq 1$. Among the k_p orbits of size s_p , o^* is forced in and the other $\mu_p - 1$ chosen from $k_p - 1$; for $i \neq p$ the μ_i orbits are any of the k_i . Given the orbit set, the bundle ranges over $\text{BUNDLE}(\mu)$ possibilities and the remainder independently over $\text{fix}(k - \mu)$. Summing over μ gives the identity. $\square$

Theorem 3 (Correctness). *Assume Theorem 2. For every cycle type λ with orbit vector k ,*

$$\text{cfix}(\lambda) = \sum_t g_1(t, 0, k), \quad \text{fix}(\lambda) = \text{Proposition 2},$$

and hence $\text{A048192}(n) = \sum_{\lambda \vdash n} \text{cfix}(\lambda)/z_\lambda$ and $\text{A048193}(n) = \sum_{\lambda \vdash n} \text{fix}(\lambda)/z_\lambda$ by (1).

Proof. By Lemma 1 the orbit-level evaporation and all eight conditions are well-defined. The four composition recurrences lift verbatim and the four single-component recurrences hold by the Lift Principle (Proposition 1) with the Component-Orbit Decomposition (Theorem 1); the recursion on the order- (N/d) sub-worlds is well-founded (Lemma 4) with base case $\pi = \text{id}$, where the functions and recurrences are those of Theorem 2. Therefore $\sum_t g_1(t, 0, k)$ counts the connected π -invariant chordal graphs and Proposition 2 counts all of them. $\square$

6 Running time

For a cycle type λ with distinct sizes $s_1 < \dots < s_r$ and multiplicities m_i , write $P(\lambda) = \prod_{i=1}^r (m_i + 1)$.

Lemma 5. *Let $r = r(\lambda)$ be the number of distinct part sizes of λ . The dynamic program for λ , including its divisor-bundle sub-instances, runs in time*

$$T(\lambda) \leq n^{O(1)} P(\lambda)^{O(1)} (n+1)^{O(r)}.$$

Proof. States. Each function is memoized over an index $t \leq n$ and between one and four tuples $(x, k, z, \text{ and } l \text{ for the } f\text{-functions})$, each bounded coordinatewise by the multiplicities, so taking at most $P(\lambda)$ values; there are $O(nP(\lambda)^4)$ states per function.

Block generating functions. A recurrence's bundle term builds a multivariate generating function (4) in $r' \leq r$ block variables of total degree at most the number of blocks $\sum_i \gcd(d, s_i)x_i \leq n$. Such a polynomial has at most $\prod_\tau (\beta_d(x)_\tau + 1) \leq (n+1)^r$ monomials, and is assembled in $(n+1)^{O(r)}$ ring operations. (This corrects a naive $n^{O(1)}$ estimate: the monomial count is exponential in the number of distinct block sub-types.)

Work per state. Besides the generating function, a recurrence sums over a sub-multiset of k ($\leq P(\lambda)$ terms; $\leq P(\lambda)^2$ for the absorption recurrence) and over divisors (at most $d(c) = n^{o(1)} \leq n$), and evaluates the sub-world g_1 at each generating-function monomial (a memo lookup). All counts are non-negative integers bounded by the number of labeled chordal graphs on n vertices, $\leq 2^{\binom{n}{2}}$, hence of $O(n^2)$ bits, so each arithmetic operation costs $\text{poly}(n)$. Work per state is therefore $n^{O(1)} P(\lambda)^2 (n+1)^r$.

Sub-instances. A bundle term calls a sub-instance whose multiplicity vector has $P(\lambda') \leq P(\lambda)$ (the map $s \mapsto s/\gcd(d, s)$ only merges parts, and $(a+1)(b+1) \geq a+b+1$) and at most r distinct sizes; the recursion depth is $\leq \log_2 n$ (each step at least halves the largest size). The number of distinct reachable sub-instances is at most $\prod_i d(s_i) \leq 2^{O(r)}$; charging each by the same bound multiplies the total by $2^{O(r)}$, which is absorbed into the $(n+1)^{O(r)}$ factor (as $(n+1)^{O(r)} \geq 2^{O(r)}$), leaving the stated form. $\square$

Lemma 6. $\max_{\lambda \vdash n} P(\lambda) \leq e^{(\sqrt{2}+o(1))\sqrt{n}} = e^{O(\sqrt{n})}$, and the number of distinct part sizes satisfies $r(\lambda) \leq \sqrt{2n}$.

Proof. The distinct sizes satisfy $s_i \geq i$, so $\sum_i i m_i \leq \sum_i s_i m_i = n$. Relaxing $m_i \geq 0$ to the reals can only increase $\max \sum_i \log(1 + m_i)$ subject to $\sum_i i m_i \leq n$. The relaxed objective is concave; a Lagrange multiplier gives $1 + m_i = 1/(\mu i)$ for $i \leq C := 1/\mu$ and $m_i = 0$ beyond, and the constraint $\sum_{i \leq C} (\frac{1}{\mu} - i) \leq n$ yields $C^2/2(1 + o(1)) \leq n$, i.e. $C \leq (\sqrt{2} + o(1))\sqrt{n}$. Then $\sum_{i \leq C} \log \frac{C}{i} = C \log C - \log(C!) = C + O(\log C)$ by Stirling, so $\log \max_\lambda P(\lambda) \leq (\sqrt{2} + o(1))\sqrt{n}$. Finally, since the s_i are distinct, $1 + 2 + \dots + r \leq \sum_i s_i \leq \sum_i s_i m_i = n$, whence $r(r+1)/2 \leq n$ and $r \leq \sqrt{2n}$. $\square$

Theorem 4 (Sub-exponential running time). *Computing $A048192(n)$ and $A048193(n)$ by the equivariant Burnside method takes time $n^{O(\sqrt{n})} = e^{O(\sqrt{n} \log n)}$.*

Proof. The total time is $\sum_{\lambda \vdash n} T(\lambda)$ plus polynomial Burnside bookkeeping. By Lemma 5, $T(\lambda) \leq n^{O(1)} P(\lambda)^{O(1)} (n+1)^{O(r)}$. By Lemma 6, $P(\lambda)^{O(1)} \leq e^{O(\sqrt{n})}$ and $r \leq \sqrt{2n}$, so $(n+1)^{O(r)} \leq (n+1)^{O(\sqrt{2n})} = e^{O(\sqrt{n} \log n)}$; hence $T(\lambda) \leq e^{O(\sqrt{n} \log n)} = n^{O(\sqrt{n})}$. The number of partitions is $p(n) = e^{\Theta(\sqrt{n})}$ (Hardy–Ramanujan), so $\sum_{\lambda \vdash n} T(\lambda) \leq p(n) \cdot \max_\lambda T(\lambda) \leq n^{O(\sqrt{n})}$. $\square$

Remark 1. The bound is sub-exponential but not polynomial: the partitions into distinct parts, of which there are up to $r = \Theta(\sqrt{n})$, force the $\sqrt{n}$ in the exponent. For families with a bounded number of distinct part sizes, $P(\lambda) = n^{O(1)}$ and the cost is polynomial. The general parameterized bound $O(2^{7\mu} n^9)$ of [2] is, for the cycle types with $\mu = \Theta(n)$ that dominate the Burnside sum, single-exponential in n ; Theorem 4 is the precise sense in which the present method is feasible across the full partition set.

7 Computational results

The recurrences of Section 5 and the Burnside sum (1) were implemented and evaluated exactly in rational arithmetic. Table 1 lists both sequences; the entries for $n \leq 15$ reproduce the previously published values, and $n = 16, 17, 18$ are new (submitted to the OEIS, pending review).

We stress that extending these sequences is not a matter of generating one more graph. The Burnside sum (1) requires the exact value of $\text{fix}(\lambda)$ for *every* cycle type $\lambda \vdash n$, and the number of cycle types is the partition number $p(n)$, which grows sub-exponentially ($p(15) = 176$, $p(18) = 385$). The difficulty is concentrated in the cycle types with many small orbits, where direct enumeration is hopeless and the equivariant recurrences of Section 5 are essential; it is computing all of these $\text{fix}(\lambda)$, not enumerating graphs, that had kept the sequences at $n = 15$.

n	A048193(n) (all)	A048192(n) (connected)
8	2119	1614
9	14524	11911
10	126758	109539
11	1392387	1247691
12	19109099	17566431
13	326005775	305310547
14	6905776799	6558690953
15	181945055235	174688164414
16	5985406996403	5796153514484
17	247178491630853	241003010628949
18	12895963060540295	12642592677074970

Table 1: Unlabeled chordal graph counts. Bold entries ($n \geq 16$) are new.

The computation was validated four independent ways.

1. *Known data.* The Burnside sum reproduces all published values of both sequences for $1 \leq n \leq 15$ exactly.
2. *Direct enumeration.* For every cycle type small enough to enumerate, $\text{fix}(\lambda)$ and $\text{cfix}(\lambda)$ from the dynamic program agree exactly with a backtracking enumerator over orbit edge-classes (which lays down all π -invariant edge-orbits and prunes on hereditary chordality). This was checked across all orbit sizes $1 \leq s \leq 16$, every single c -cycle, and many mixed and composite types; a representative sample appears in Table 2.
3. *Labeled limit.* $\text{fix}(1^n)$ reproduces the labeled chordal totals [A058862](#)(n) for $1 \leq n \leq 10$.
4. *Euler transform.* Chordal graphs are closed under disjoint union, so the total count is the Euler transform of the connected count; this identity holds for the computed values through $n = 18$, so the two independently assembled sequences are mutually consistent at the new terms.

By Theorem 4 the method extends to larger n in sub-exponential time; the values above were obtained on a single processor.

Data availability. The two computed tables ($n = 1, \dots, 18$) have SHA-256 digests

A048192 (connected)	e91c9a6d936007b4814503593fa5522d265eb254414ae484815a5b6e33a625e6
A048193 (all)	13f8305370f6dafdef2756c86ff3847a4ffc0d4f9138217ca3db8b087c111298

λ	n	$\text{fix}(\lambda)$	$\text{cfix}(\lambda)$	brute
1^3	3	8	4	✓
21^2	4	15	7	✓
2^2	4	13	7	✓
31^2	5	15	7	✓
321	6	26	11	✓
3^2	6	22	15	✓
41^2	6	22	10	✓
421	7	69	37	✓
4^2	8	64	37	✓
63	9	37	23	✓
3^3	9	897	745	✓
84	12	92	52	✓
93	12	34	22	✓
105	15	113	95	✓

Table 2: Per-cycle-type values from the dynamic program, each agreeing exactly with direct backtracking enumeration. The composite sizes 4, 6, 8, 9, 10 and the cross-divisor types (6 3, 8 4, 9 3, 10 5) exercise the divisor-bundle recursion.

The implementation (the recurrences of Appendix A, the brute-force oracle, the verification scripts, the b-files, and this paper’s source) is publicly available at github.com/msun170/chordal-counting.

8 Conclusion

We have given a concrete equivariant dynamic program for counting permutation-invariant chordal graphs, built on the labeled evaporation program of [1]. Its correctness rests on a divisor-bundle decomposition (Theorem 1) that reduces each equivariant recurrence to its labeled counterpart, and the full Burnside computation runs in sub-exponential time (Theorem 4). The resulting enumeration extends the unlabeled chordal sequences A048192 and A048193 past their previous frontier. The orbit-bundle lifting strategy may be adaptable to other graph classes that admit a compatible labeled decomposition recurrence; we leave such extensions to future work.

A The explicit equivariant recurrences

We collect the eight recurrences over the state variables of Section 5.1. Vectors are indexed by the sizes $S = (s_1 < \cdots < s_r)$; $\binom{k}{m} := \prod_i \binom{k_i}{m_i}$; $p = p(k)$ is the largest index with $k_p > 0$; and the marking coefficient is

$$M(k, \kappa) = \binom{k_p - 1}{\kappa_p - 1} \prod_{i \neq p} \binom{k_i}{\kappa_i}.$$

A bundle of period d absorbs only sizes divisible by d ; we write $\kappa \triangleright d$ for “ $\kappa_i > 0 \Rightarrow d \mid s_i$,” and κ/d for the vector with $s_i \mapsto s_i/d$. The boundary terms use the block generating functions $\Gamma_d, \Gamma_d^{\text{cov}}$ of (4); $[\Phi]_a$ is the coefficient of $\prod_\tau v_\tau^{a_\tau}$, and $g_1^{(N/d)}(t, a, \nu)$ is the sub-world function with block-boundary a . Define the three peeled-component operators (the $d = 1$ summand A and the

$d > 1$ summands B_d , for the three boundary conditions $\Phi \in \{\text{see}, \text{not-all}, \text{cover}\}$):

$$\begin{aligned}
A^{\text{see}}(\kappa) &= \sum_{0 \neq x' \leq x} \left(\binom{x}{x'} - \binom{z}{x'} \right) g_1(t, x', \kappa), \\
A^{\text{not-all}}(\kappa) &= \sum_{0 \neq x' \leq x} \left(\binom{x}{x'} - \binom{z}{x'} \right) g_1(t, x', \kappa), \\
B_d^{\text{see}}(\kappa) &= d^{|\kappa|-1} \sum_a [\Gamma_d(x) - \Gamma_d(z)]_a g_1^{\langle N/d \rangle}(t, a, \kappa/d), \\
B_d^{\text{cov}}(\kappa) &= d^{|\kappa|-1} \sum_a [\Gamma_d^{\text{cov}}(x)]_a g_1^{\langle N/d \rangle}(t, a, \kappa/d), \\
B_d^{\text{not-all}}(\kappa) &= d^{|\kappa|-1} \sum_a [\Gamma_d(x) - \Gamma_d^{\text{cov}}(x) - \Gamma_d(z)]_a g_1^{\langle N/d \rangle}(t, a, \kappa/d) \quad (0 \text{ if } z = x).
\end{aligned}$$

Base cases and invalid states. Every function returns 0 outside its domain and on impossible states: $g_1 = 0$ if $t \leq 0$ or $k = 0$; $g(0, x, k, z) = [k = 0]$ and $g = 0$ for $t < 0$; $\tilde{g}(t, x, 0, z) = 1$; $\hat{g}(t, x, 0, z) = 1$ and $\hat{g} = 0$ for $t < 1$; $g_2 = 0$ if $t \leq 0$ or $k = 0$; $f = 0$ for $t \leq 0$, $f(1, x, l, k) = [k = 0]$, and $f = 0$ for $k = 0$, $t \geq 2$; and $\tilde{f} = \hat{f} = 0$ for $t \leq 1$ or $k = 0$. Empty sums are 0 and empty products 1; $\binom{a}{b} = 0$ for $b > a$ or $b < 0$, so a sub-multiset that does not fit contributes nothing, as does any state with a negative coordinate. A clique-number bound ω is carried with the recursion, and $f = 0$ once $\sum_i s_i(x_i + l_i) > \omega$. The recursion is finite: t strictly decreases through $f \rightarrow g \rightarrow \tilde{g}$, and the world order $N/d < N$ strictly decreases at each bundle step (Lemma 4). Then:

$$g_1(t, x, k) = \sum_{0 \neq \ell \leq k} \binom{k}{\ell} f(t, x, \ell, k - \ell), \quad (\text{R1})$$

$$f(t, x, l, k) = \sum_{m \leq k} \binom{k}{m} \tilde{f}(t, x, l, m) g(t - 2, x + l, k - m, x), \quad (\text{R2})$$

$$g(t, x, k, z) = \sum_{m \leq k} \binom{k}{m} \tilde{g}(t, x, m, z) g(t - 1, x, k - m, z), \quad (\text{R3})$$

$$\tilde{g}(t, x, k, z) = \sum_{\kappa_p \geq 1} M(k, \kappa) \left[A^{\text{see}}(\kappa) + \sum_{d > 1, \kappa \triangleright d} B_d^{\text{see}}(\kappa) \right] \tilde{g}(t, x, k - \kappa, z), \quad (\text{R4})$$

$$\hat{g}(t, x, k, z) = \sum_{\kappa_p \geq 1} M(k, \kappa) \left[A^{\text{not-all}}(\kappa) + \sum_{d > 1, \kappa \triangleright d} B_d^{\text{not-all}}(\kappa) \right] \hat{g}(t, x, k - \kappa, z), \quad (\text{R5})$$

$$g_2(t, x, k) = \sum_{\kappa_p \geq 1} M(k, \kappa) \left[g_1(t, x, \kappa) R(k - \kappa) + \sum_{d > 1, \kappa \triangleright d} B_d^{\text{cov}}(\kappa) R_0(k - \kappa) \right], \quad (\text{R6})$$

$$\begin{aligned}
\tilde{f}(t, x, l, k) &= \hat{f}(t, x, l, k) \\
&+ \sum_{0 \neq m \leq k} \binom{k}{m} \left[g_1(t - 1, x + l, m) \hat{f}(t, x, l, k - m) + g_2(t - 1, x + l, m) \hat{g}(t - 1, x + l, k - m, x) \right].
\end{aligned} \quad (\text{R7})$$

where in (R6) $R(\cdot) = g_1 + g_2$ and $R_0(m) = [m = 0] + R(m)$. Finally the absorption recurrence: with $\Gamma_d^{\text{tou}}(\sigma) = \prod_i ((1 + v_{\tau_i})^{\text{gcd}(d, s_i)} - 1)^{\sigma_i}$ the “touched-orbit” generating function, the weight $w(xp, lp) = ((\binom{x}{xp} - [lp=0]\binom{z}{xp}))\binom{l}{lp}$, and the successor $\text{nx}_{lp}(k') = \hat{g}(t - 1, x + lp, k', z)$ if $lp = l$ and

$\hat{f}(t, x+lp, z, l-lp, k')$ otherwise,

$$\hat{f}(t, x, z, l, k) = \sum_{\kappa_p \geq 1} M(k, \kappa) \sum_{\substack{xp \leq x, lp \leq l \\ (xp, lp) \notin \{0, (x, l)\}}} w(xp, lp) \text{nx}_{lp}(k - \kappa) \Theta_\kappa(xp + lp), \quad (\text{R8})$$

where the touched-component value, the absorption analogue of the bracket in (R4), is

$$\Theta_\kappa(\sigma) = g_1(t-1, \sigma, \kappa) + \sum_{d > 1, \kappa \triangleright d} d^{|\kappa|-1} \sum_a [\Gamma_d^{\text{tou}}(\sigma)]_a g_1^{\langle N/d \rangle}(t-1, a, \kappa/d).$$

Setting all sizes to 1 ($r = 1$, $N = 1$, no $d > 1$) recovers the labeled recurrences of Theorem 2.

B A worked example: $\lambda = 2^2$

Let $\pi = (1\ 2)(3\ 4)$, so $N = 2$ and the only sizes are $\{2\}$. We compute $\text{fix}(2^2)$ from Proposition 2 and Lemma 2, displaying the divisor-bundle terms.

A single component-bundle on j of the 2-cycles has, by Theorem 1,

$$\text{BUNDLE}(2^j) = \underbrace{\text{cfix}(2^j)}_{d=1} + \underbrace{2^{j-1} \text{cfix}^{(1)}(1^j)}_{d=2}, \quad \text{cfix}^{(1)}(1^j) = \text{A007134}(j),$$

the $d = 2$ term being a swapped pair of labeled components. With $\text{cfix}(2^1) = 1$, $\text{cfix}(2^2) = 7$ and $\text{A007134}(1) = \text{A007134}(2) = 1$,

$$\text{BUNDLE}(2^1) = 1 + 2^0 \cdot 1 = 2, \quad \text{BUNDLE}(2^2) = 7 + 2^1 \cdot 1 = 9.$$

Proposition 2 with $k = (2)$ (marked size 2) gives, over $\mu \in \{(1), (2)\}$,

$$\text{fix}(2^2) = \binom{1}{0} \text{BUNDLE}(2^1) \text{fix}(2^1) + \binom{1}{1} \text{BUNDLE}(2^2) \text{fix}(\emptyset).$$

Here $\text{fix}(2^1) = \text{BUNDLE}(2^1) = 2$ and $\text{fix}(\emptyset) = 1$, so

$$\text{fix}(2^2) = 1 \cdot 2 \cdot 2 + 1 \cdot 9 \cdot 1 = 13,$$

matching the direct enumeration in Table 2. The contributions $2^0 \cdot 1$ and $2^1 \cdot 1$ are exactly the period-2 bundles (an edge between the two cyclically-paired vertices, in its 2^{j-1} alignments), which the $d = 1$ count alone would miss.

References

- [1] U. Hébert-Johnson, D. Lokshtanov, E. Vigoda. *Counting and Sampling Labeled Chordal Graphs in Polynomial Time*. arXiv:2308.09703, 2023.
- [2] U. Hébert-Johnson, D. Lokshtanov. *Sampling Unlabeled Chordal Graphs in Expected Polynomial Time*. STACS 2025; arXiv:2501.05024.